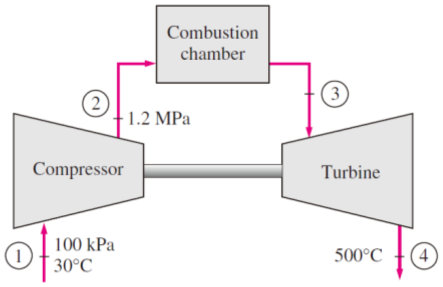
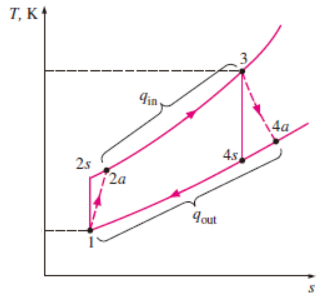
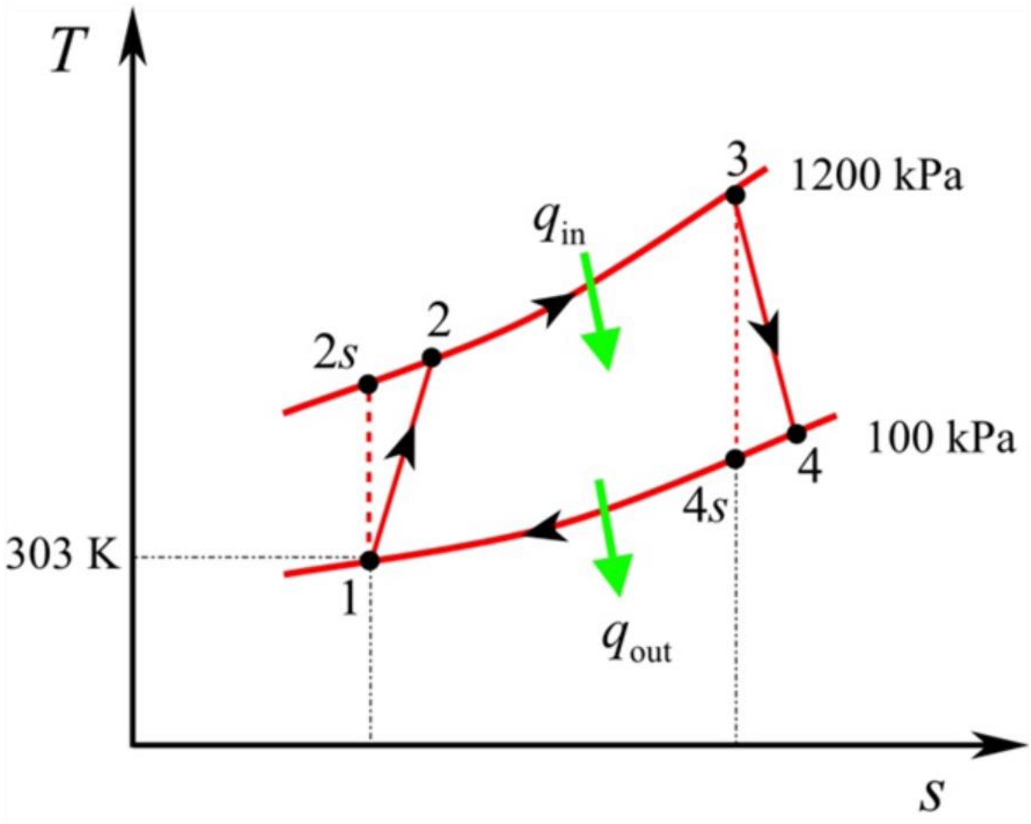
2 percent and a turbine isentropic efficiency ($\eta_t=(h_3-h_{4a})/(h_3-h_{4s})$) of 88 percent, determine (a) the net power output, (b) the back work ratio, and (c) the thermal efficiency. （给求解过程）





Draw the T-s diagram for the system as follows:



Take the value of enthalpy from "Ideal-gas properties of air" at temperature $T_1 = 303\ \text{K}$ and $p_1 = 100\,\text{kPa}$.

$h_1 = \mathbf{303.5}\ \text{kJ/kg}$
$s_1 = \mathbf{6.881}\ \text{kJ/kg}\cdot\text{K}$

Consider the isentropic process 1-2s:

Obtain the value of enthalpy at $s_1 = \mathbf{6.881}\ \text{kJ/kg}\cdot\text{K}$ and $p_2 = 1200\,\text{kPa}$ from "Ideal-gas properties of air" take the value of enthalpy at relative pressure.

$h_{2s} = \mathbf{617.92}\ \text{kJ/kg}$

Calculate the enthalpy at state 2 by using the isentropic efficiency of the compressor equation.

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1}$$

Substitute $\mathbf{303.5}\ \text{kJ/kg}$ for h_1, $\mathbf{617.92}\ \text{kJ/kg}$ for h_{2s}, and 0.82 for η_C.

$$0.82 = \frac{\mathbf{617.92 - 303.5}}{h_2 - \mathbf{303.5}}$$
$$0.82h_2 - \mathbf{248.87} = \mathbf{314.42}$$
$$h_2 = \mathbf{686.94}\ \text{kJ/kg}$$

Consider the process 3-4: expansion

$T_4 = 500°\text{C}$
$h_4 = \mathbf{793.12}\ \text{kJ/kg}$

Calculate the enthalpy at state 3 by using the isentropic efficiency of the compressor equation.

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}}$$

Substitute $\mathbf{793.12}\ \text{kJ/kg}$ for h_4, and 0.88 for η_T.

$$0.88 = \frac{h_3 - \mathbf{793.12}}{h_3 - h_{4s}} \quad \text{...... (1)}$$

Find the value of h_3 by trial and error method as follows:

Assume h_3 =**1400** kJ/kg

Obtain the entropy value at h_3 =**1400** kJ/kg and $p = 1200\,\text{kPa}$

$s_3 = \mathbf{7.7313}\,\text{kJ/kg}\cdot\text{K}$

Consider the process 3-4s: isentropic expansion

For isentropic process

$s_3 = s_4$
$s_4 = \mathbf{7.7313}\,\text{kJ/kg}\cdot\text{K}$

Obtain the enthalpy value at $s_4 = \mathbf{7.7313}\,\text{kJ/kg}\cdot\text{K}$ and $p = 100\,\text{kPa}$

$h_{4s} = \mathbf{706}\,\text{kJ/kg}\cdot\text{K}$

Substitute **1400** kJ/kg for h_3, and **706** kJ/kg·K for h_{4s} in equation (1).

$$\mathbf{0.87} = \frac{\mathbf{1400 - 793.12}}{\mathbf{1400 - 706}}$$
$$0.88 \neq \mathbf{0.87}$$

This is close value so assume another h_3 value close to **1400** kJ/kg.

Assume h_3 =**1410** kJ/kg

Obtain the entropy value at h_3 =**1410** kJ/kg and $p = 1200\,\text{kPa}$

$s_3 = \mathbf{7.739}\,\text{kJ/kg}\cdot\text{K}$

Consider the process 3-4s: isentropic expansion

For isentropic process

$s_3 = s_4$
$s_4 = \mathbf{7.739}\,\text{kJ/kg}\cdot\text{K}$

Obtain the enthalpy value at $s_4 = \mathbf{7.739}\,\text{kJ/kg}\cdot\text{K}$ and $p = 100\,\text{kPa}$

$h_{4s} = \mathbf{712}\,\text{kJ/kg}\cdot\text{K}$

Substitute **1410** kJ/kg for h_3, and **712** kJ/kg·K for h_{4s} in equation (1).

$$0.88 = \frac{\mathbf{1410 - 793.12}}{\mathbf{1410 - 712}}$$

So, consider the value of enthalpy at state-3 is **1410** kJ/kg.

Calculate the mass flow rate as follows:

$$\dot{m} = \frac{p_1 \dot{V}_1}{RT_1}$$

Substitute $100\,\text{kPa}$ for p_1, $\frac{150}{60}\,\text{m}^3/\text{s}$ $\dot{V}_1$, and $30 + 273 = 303\,\text{K}$ for T_1.

$$\dot{m} = \frac{(100)\left(\frac{150}{60}\right)}{(0.287)(303)}$$
$$= 2.87\,\text{kg/s}$$

(a)

Calculate the compressor work (W_{comp}) as follows:

$$W_{comp} = \dot{m}(h_2 - h_1)$$

Substitute $2.87\,\text{kg/s}$ for $\dot{m}$, $\mathbf{686.94}\,\text{kJ/kg}$ for h_2, and $\mathbf{303.5}\,\text{kJ/kg}$ for h_1.

$$W_{comp} = 2.87(\mathbf{686.94} - \mathbf{303.5})$$
$$= \mathbf{1100.5}\,\text{kW}$$

Calculate the turbine work $(W_{Turbine})$ as follows:

$$W_{Turbine} = \dot{m}(h_3 - h_4)$$

Substitute $2.87\,\text{kg/s}$ for $\dot{m}$, $\mathbf{1410}\,\text{kJ/kg}$ for h_3, and $\mathbf{793.12}\,\text{kJ/kg}$ for h_4.

$$W_{Turbine} = 2.87(\mathbf{1410} - \mathbf{793.12})$$
$$= \mathbf{1770.4}\ \text{kW}$$

Calculate the net work done as follows:

$$W_{net} = W_{Turbine} - W_{comp}$$

Substitute $\mathbf{1101}$ kW for W_{comp}, and $\mathbf{1770.4}$ kW for $W_{Turbine}$.

$$W_{net} = \mathbf{1770.4} - \mathbf{1101}$$
$$= \mathbf{669.4}\ \text{kW}$$

Therefore, net work done of the system is $\boxed{\mathbf{669.4}}$.

(b)

Calculate the back work ratio as follows:

$$r_{bw} = \frac{W_{comp}}{W_{Turbine}}$$

Substitute $\mathbf{1100.5}$ kW for W_{comp}, and $\mathbf{1770.4}$ kW for $W_{Turbine}$.

$$r_{bw} = \frac{\mathbf{1100.5}}{\mathbf{1770.4}} = \mathbf{0.622}$$

Therefore, back work ratio is $\boxed{\mathbf{0.622}}$

(c)

Calculate the thermal efficiency as follows:

$$\eta_{th} = \frac{W_{net}}{q_{in}}$$

Calculate the heat addition as follows:

$$q_{in} = m\left(h_3 - h_2\right)$$

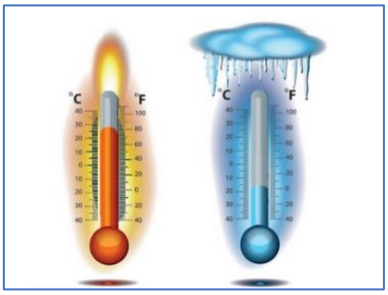
Substitute $2.87\,\text{kg/s}$ for $\dot{m}$, $\mathbf{1410}\,\text{kJ/kg}$ for h_3, and $\mathbf{686.94}\,\text{kJ/kg}$ for h_2.

$$\dot{q}_{in} = 2.87\left(\mathbf{1410} - \mathbf{687}\right) = \mathbf{2075}\,\text{kW}$$

Substitute $\mathbf{669.4}$ kW for W_{net}, and $\mathbf{2075}$ kW for $\dot{q}_{in}$.

$$\eta_{th} = \frac{\mathbf{669.4}}{\mathbf{2075}} = \mathbf{0.323}$$

Therefore, back work ratio is $\boxed{\mathbf{0.323}}$



2. Whether the temperature scale of mercurial thermometer should be uniform or not? Please does derivation considering the coefficient of *linear thermal expansion* (α) is constant: is the volume increment of mercury caused by temperature rise linearly related to the temperature difference? Assuming the length of the material is L_0, the length increases ΔL when temperature rises ΔT. Then α can be calculated by follows. （给求解过程）

$$\alpha = \frac{\Delta L}{L_0 \Delta T}$$

Not uniform.

$$\Delta L = \alpha L_0 \Delta T$$

$$\Delta V = \left(L_0 + \Delta L\right)^3 - L_0^3 = 3L_0^2\Delta L + 3L_0\Delta L^2 + \Delta L^3 = 3\alpha L_0^3\Delta T + 3\alpha^2 L_0^3\Delta T^2 + \alpha^3 L_0^3\Delta T^3$$

So, when the linear thermal expansion is very small, the second and third terms on the right-hand side of the equal sign can be ignored. The volume increment of mercury caused by temperature rise is linearly related to the temperature difference.

4. One mole of an ideal gas at 25°C and 1 atm undergoes the following reversibly conducted cycle:

a. An isothermal expansion to 0.5 atm, followed by

b. An isobaric expansion to 100°C, followed by

c. An isothermal compression to 1 atm, followed by

d. An isobaric compression to 25°C.

The system then undergoes the following reversible cyclic process.

a. An isobaric expansion to 100°C, followed by

b. A decrease in pressure at constant volume to the pressure ***P*** atm, followed by

c. An isobaric compression at ***P*** atm to 24.5 liters, followed by

d. An increase in pressure at constant volume to 1 atm.

Calculate the value of ***P*** which makes the work done on the gas during the first cycle equal to the work done by the gas during the second cycle. （给求解过程）

$$w_{1,a} = nRT_{1,a}\ln\left(\frac{P_{1,0}}{P_{1,a}}\right) = 1\times 8.314\times 298.15\times\ln\left(\frac{1}{0.5}\right) = 1718J$$

$$w_{1,b} = P_{1,b}\left(V_{1,b} - V_{1,a}\right) = nRT_{1,b} - nRT_{1,a} = 1\times 8.314\times\left(100-25\right) = 624J$$

$$w_{1,c} = nRT_{1,c}\ln\left(\frac{P_{1,b}}{P_{1,c}}\right) = 1\times 8.314\times 373.15\times\ln\left(\frac{0.5}{1}\right) = -2150J$$

$$w_{1,d} = P_{1,d}\left(V_{1,d} - V_{1,c}\right) = nRT_{1,d} - nRT_{1,c} = 1\times 8.314\times\left(25-100\right) = -624J$$

$$w_1 = w_{1,a} + w_{1,b} + w_{1,c} + w_{1,d} = 1718 + 624 - 2150 - 624 = -432J$$

$$w_{2,a} = P_{2,a}\left(V_{2,a} - V_{1,d}\right) = nRT_{2,a} - nRT_{1,d} = 1\times 8.314\times\left(100-25\right) = 624J$$

$$w_{2,b} = 0$$

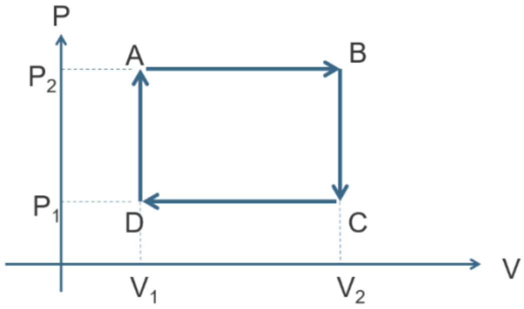
$$w_{2,c} = P_{2,c}\left(V_{2,c} - V_{2,b}\right) = P_{2,c}\left(V_{2,c} - \frac{nRT_{2,a}}{P_{2,a}}\right) = P\left(24.5\times 10^{-3} - \frac{1\times 8.314\times 373.15}{101325}\right) = -6.12\times 10^{-3}P$$

$$w_{2,b} = 0$$

$$w_2 = w_{2,a} + w_{2,b} + w_{2,c} + w_{2,d} = 624 + 0 - 6.12\times 10^{-3}P + 0 = -w_1 = 432J$$

$$P = 31373Pa = 0.3atm$$

3. One mole of an ideal gas are taken round the following cycle. Please find the efficiency of the cycle. (给答案))



A → B: Constant Pressure Process
B → C: Constant Volume Process
C → D: Constant Pressure Process
D → A: Constant Volume Process

$$w=(P_2-P_1)(V_2-V_1)$$

$$q=C_p(T_B-T_A)+C_v(T_A-T_D)=C_p\left(\frac{P_2V_2}{R}-\frac{P_2V_1}{R}\right)+C_v\left(\frac{P_2V_1}{R}-\frac{P_1V_1}{R}\right)$$

$$=P_2(V_2-V_1)+\frac{C_v}{R}(P_2V_2-P_1V_1)$$

$$\eta=\frac{w}{q}=\frac{(P_2-P_1)(V_2-V_1)}{P_2(V_2-V_1)+\frac{C_v}{R}(P_2V_2-P_1V_1)}$$

1. To ease the energy crisis, the idea of an air-powered car has been mentioned repeatedly. Assuming the current efficiency of gasoline engine is about 36%, if a steel cylinder that can withstand pressure of 100 atm is used in the air-powered car, what size of the steel cylinder is needed to make the air-powered car run at the same distance as gasoline engine car with full tank (24 liters). Assuming the efficiency of air-powered car is 100%. Ignore the mass of the steel cylinder. (The calorific value of gasoline is 33580 kJ/liter. The ambient temperature is 300K.) （给数字答案）

$$w_{total}=36\%\times 24\times 33580=290131.2kJ$$

$$\phi=RT_0\left(\frac{P_0}{P_1}-1\right)+RT_0\ln\frac{P_1}{P_0}=RT_0\left(\ln\frac{P_1}{P_0}+\frac{P_0}{P_1}-1\right)$$

$$\phi=\frac{8.314}{28.97}\times 300\times\left(\ln\frac{10^7}{10^5}+\frac{10^5}{10^7}-1\right)=311\ kJ/kg$$

$$m=\frac{w_{total}}{\phi}=933kg$$

$$V=\frac{m}{28.97}\frac{RT}{P}=8m^3$$